Task-17 :- Requirement Analysis

1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for designing and implementing a **Multi-VPC AWS network environment across two regions** for ABC Technologies.

This document outlines the architecture, system requirements, functional requirements, and non-functional requirements necessary to implement a scalable and structured cloud network.

The intended audience of this document includes:

- DevOps Engineers
- Cloud Engineers
- Network Architects
- System Administrators
- Project Stakeholders

This document will help stakeholders understand the design, connectivity model, and deployment approach for the multi-region AWS network architecture.

1.2 Scope

The scope of this project includes designing and implementing a **multi-region AWS network architecture using multiple VPCs and VPC Peering connections**.

Included in Scope

- Designing network architecture for **6 VPC environments**
- Creating **VPCs in two AWS regions**
- Implementing **advanced subnetting based on host requirements**
- Creating **public and private subnets**
- Configuring **Internet Gateways**
- Establishing **cross-region VPC peering**
- Establishing **intra-region VPC peering**
- Deploying **EC2 instances for network testing**
- Configuring **route tables and security groups**

- Verifying connectivity between networks

Not Included in Scope

- Production application deployment
- Database clustering or replication
- VPN or Direct Connect integration
- Transit Gateway configuration
- Enterprise monitoring solutions

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
AWS	Amazon Web Services
VPC	Virtual Private Cloud
CIDR	Classless Inter-Domain Routing
EC2	Elastic Compute Cloud
IGW	Internet Gateway
AZ	Availability Zone
NAT	Network Address Translation
VPC Peering	Direct network connection between two VPCs
Subnet	Logical subdivision of a VPC network
Bastion Host	Server used to securely access private resources
Route Table	Routing rules that control network traffic

1.4 References

The following resources were used during the preparation of this document:

- AWS VPC Documentation
  https://docs.aws.amazon.com/vpc/
- AWS VPC Peering Documentation
  https://docs.aws.amazon.com/vpc/latest/peering/
- AWS Networking Best Practices
  AWS Architecture Center
- AWS EC2 Documentation
  https://docs.aws.amazon.com/ec2/

1.5 Overview

This document is structured into several sections:

Section	Description
Introduction	Overview and purpose of the document
Project Overview	Project objectives and background
Functional Requirements	System functionality and use cases
Non-Functional Requirements	Performance, security, reliability
Assumptions & Dependencies	Conditions affecting implementation
Risk Analysis	Potential risks and mitigation strategies
Glossary	Explanation of key terms
Appendices	Supporting documentation

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2. Project Overview

2.1 Project Objectives

The primary objectives of this project are:

- Design a **multi-VPC AWS architecture across multiple regions**
- Implement **advanced subnetting for host requirements**

- Deploy infrastructure in **two AWS regions**
- Establish **secure connectivity between VPCs using VPC Peering**
- Implement **public and private subnet architecture**
- Deploy EC2 instances for network validation and testing
- Demonstrate structured and scalable cloud networking practices

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2.2 Business Goals

The business goals of this project include:

- Improving **network scalability and segmentation**
- Ensuring **secure communication between multiple office environments**
- Implementing **multi-region infrastructure**
- Supporting future **application expansion**
- Creating a **structured network design for enterprise cloud environments**

2.3 Background

ABC Technologies is expanding its cloud infrastructure to support multiple headquarter environments and branch office networks.

Previously, the company relied on a **single VPC network**, which limited scalability and network isolation.

To support growing business operations, the organization plans to implement a **multi-VPC architecture across two AWS regions**. This architecture will separate different business environments while allowing controlled communication between them through **VPC Peering connections**.

This approach improves network segmentation, scalability, and security.

2.4 Constraints

Constraint	Description
Time	The task must be completed within **3 hours**

Technology	Implementation must use **AWS cloud services**
Regions	Must use **us-east-1 and us-east-2 regions**
Connectivity	Must use **VPC Peering instead of Transit Gateway**
Network Design	CIDR blocks must **not overlap between VPCs**

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3. Functional Requirements

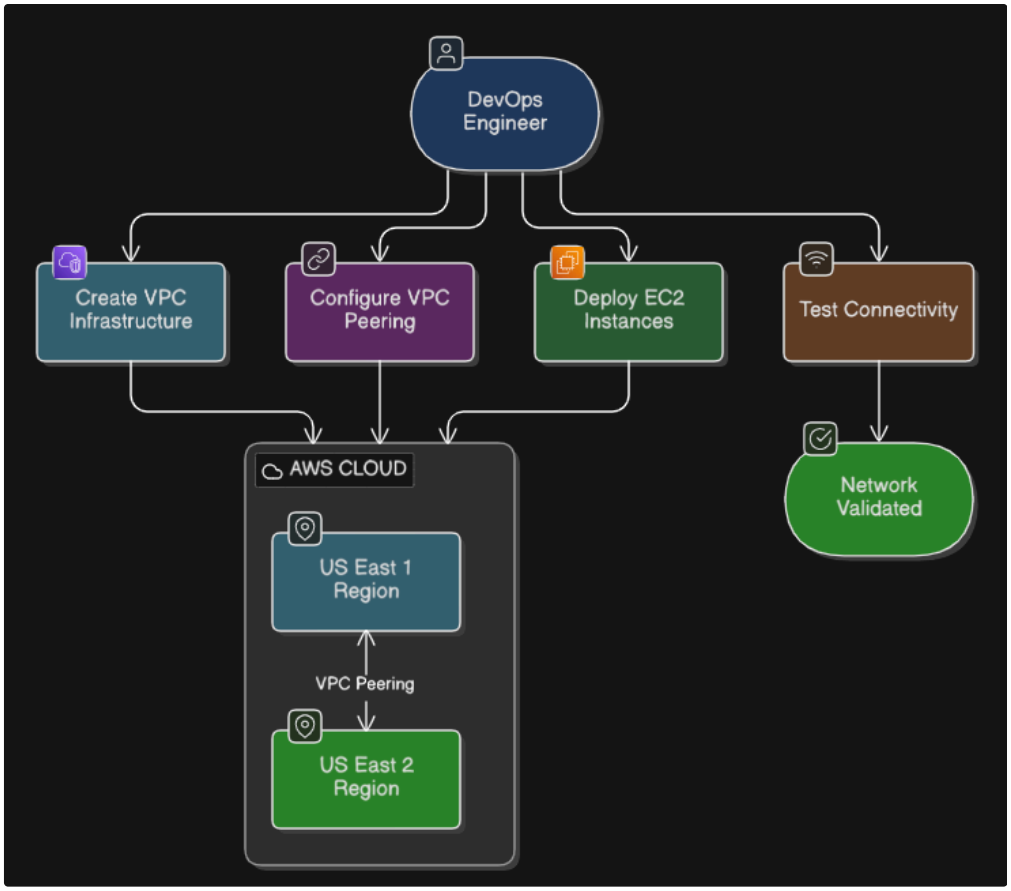
3.1 Use Case Diagram

Actors involved in the system:

- DevOps Engineer
- AWS VPC Infrastructure
- EC2 Instances
- Network Routing System

Suggested diagram structure:

```
                DevOps Engineer
                      |
                      v
              AWS VPC Infrastructure
                      |
         ----------------------------------
         |                                |
      HQ VPCs                        Office VPCs
         |                                |
         -----------VPC Peering-----------
                      |
                      v
                 EC2 Instances
                      |
                      v
              Connectivity Testing
```



3.2 Use Cases

3.2.1 Use Case 1: Create VPC Infrastructure

Description:

The DevOps engineer creates multiple VPC networks across two AWS regions.

Actors:

DevOps Engineer

Preconditions:

- AWS account access
- VPC service available
- CIDR planning completed

Postconditions:

- All required VPC networks are created.

Flow of Events:

1. DevOps engineer logs into AWS Console.

2. Creates VPC networks in specified regions.
3. Assigns CIDR ranges to each VPC.
4. Creates public and private subnets.
5. Configures internet gateways and route tables.

Alternative Flows:

- CIDR overlap occurs → VPC design must be updated.
- Incorrect subnet allocation → subnet configuration adjusted.

3.2.2 Use Case 2: Configure VPC Peering

Description:
Establish connectivity between VPC networks.

Actors:
DevOps Engineer

Preconditions:

- VPC networks already created
- CIDR ranges do not overlap

Postconditions:

- VPCs communicate via private IP addresses.

Flow of Events:

1. Create VPC peering request.
2. Accept peering connection.
3. Update route tables for both VPCs.
4. Verify network connectivity.

Alternative Flows:

- Peering request rejected.
- Route table misconfiguration blocks communication.

3.2.3 Use Case 3: Deploy EC2 Instances for Testing

Description:
Deploy EC2 instances to validate connectivity between networks.

Actors:

DevOps Engineer

Preconditions:

- VPC networks configured
- Subnets created
- Security groups configured

Postconditions:

- Instances are deployed and connectivity is verified.

Flow of Events:

1. Launch EC2 instance in public subnet.
2. Launch EC2 instance in private subnet.
3. Configure SSH access.
4. Perform connectivity tests.

Alternative Flows:

- Security group blocks traffic.
- Route table misconfiguration prevents connectivity.

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4. Non-Functional Requirements

4.1 Performance Requirements

- Network latency between VPCs should remain minimal.
- Connectivity between EC2 instances should respond quickly.
- Subnet design should support required host capacity.

4.2 Security Requirements

- Private subnet instances should not have direct internet access.
- Access to private instances should use **bastion hosts**.
- Security groups should restrict unnecessary traffic.
- Only required ports should be open.

4.3 Usability Requirements

- Infrastructure should be manageable through AWS Console.
- Network architecture should be easy to maintain.
- Documentation should clearly describe all network components.

4.4 Reliability Requirements

- Multi-region architecture improves availability.
- Network segmentation improves fault isolation.
- VPC peering ensures reliable communication between networks.

4.5 Compliance Requirements

The system should follow:

- AWS networking best practices
- Cloud infrastructure security guidelines
- Access control and least-privilege principles

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5. Assumptions and Dependencies

5.1 Assumptions

- AWS account permissions are available.
- Required AWS services are supported in selected regions.
- Internet connectivity is available for instance access.

5.2 Dependencies

The system depends on:

- AWS EC2 Service
- AWS VPC Service
- AWS VPC Peering
- Internet Gateway configuration
- Security Group rules

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6. Risk Analysis

6.1 Risk Identification

Risk	Description
CIDR overlap	Overlapping IP ranges prevent VPC peering
Routing issues	Misconfigured route tables block connectivity
Security misconfiguration	Incorrect security group rules
Connectivity failure	VPC peering not properly configured

6.2 Risk Mitigation

Risk	Mitigation Strategy
CIDR overlap	Plan CIDR ranges carefully
Routing issues	Verify route tables after configuration
Security risks	Apply minimal required security rules
Connectivity failures	Perform EC2 connectivity testing

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7. Glossary

Term	Meaning
VPC	Private network within AWS
Subnet	Logical network division within VPC
CIDR	IP address allocation format
VPC Peering	Network connection between VPCs

Bastion Host	Secure access gateway for private instances

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8. Appendices

The following supporting materials should be included:

- Network architecture diagram
- CIDR planning table
- VPC creation screenshots
- Subnet configuration screenshots
- VPC peering configuration screenshots
- Route table configuration screenshots
- EC2 instance deployment screenshots
- Connectivity test results

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9. Customer Acknowledgement

Name: _____

Title: _____

Signature: _____

Date: _____